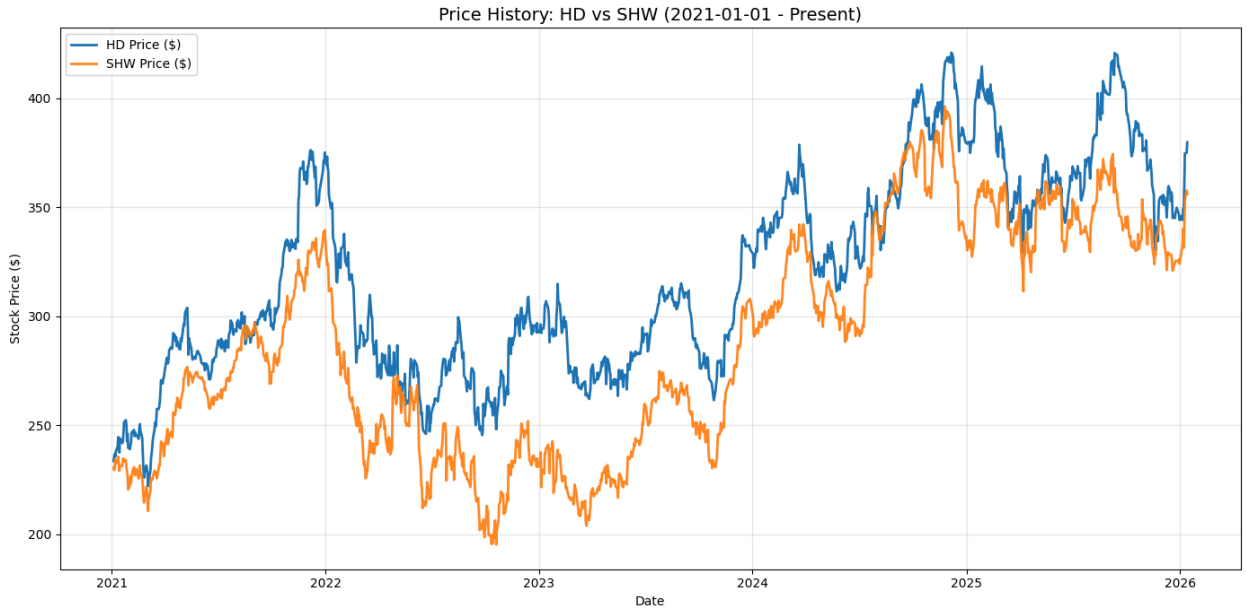
Background
Sherwin-Williams (SHW) and Home Depot (HD) are two major players in the home improvement industry. SHW specializes in their own brand of premium paint where HD provides a wider variety of products, including other brands of their own paint. HD also offers a color-matching service where they will match the SHW paint colors for customers. Painting is generally one of the more in-demand home improvements due to its simplicity.

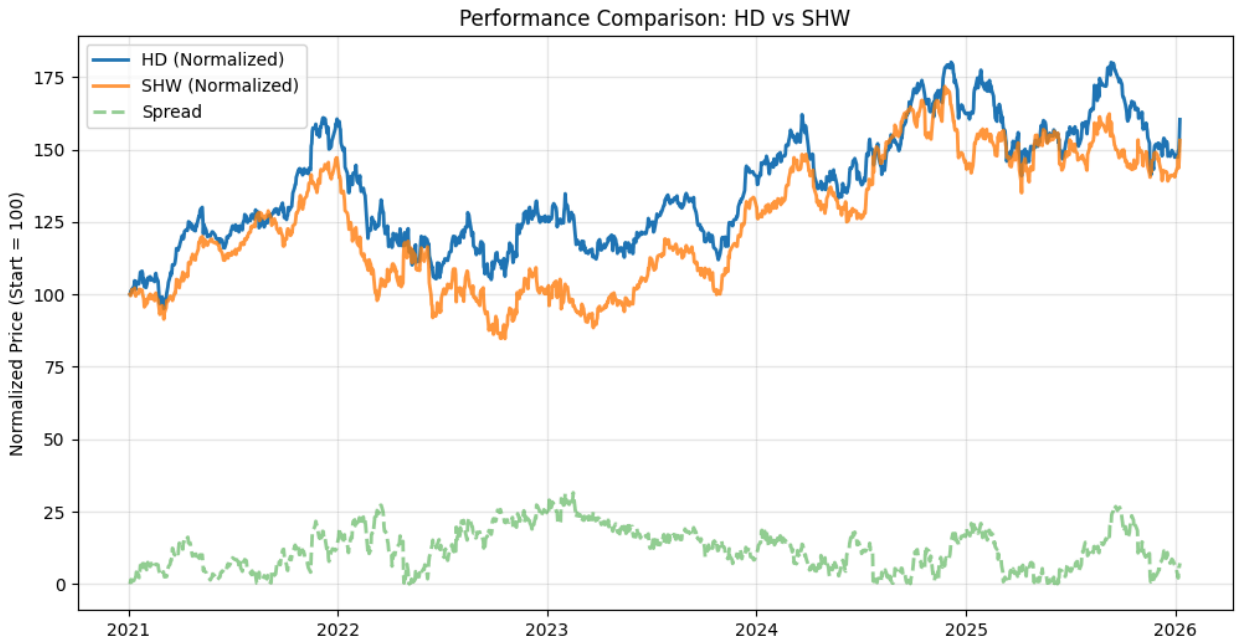
Thesis
Since both SHW and HD are two of the largest players within the home improvement and paint/coating industries, they are affected by similar trends in the housing market. They should move similarly relative to the market because customers are mainly purchasing either SHW paint or HD paint. Therefore, when one company begins to outperform the other relative to the market, there is alpha in betting on mean reversion for the spread of the pair.

EDA



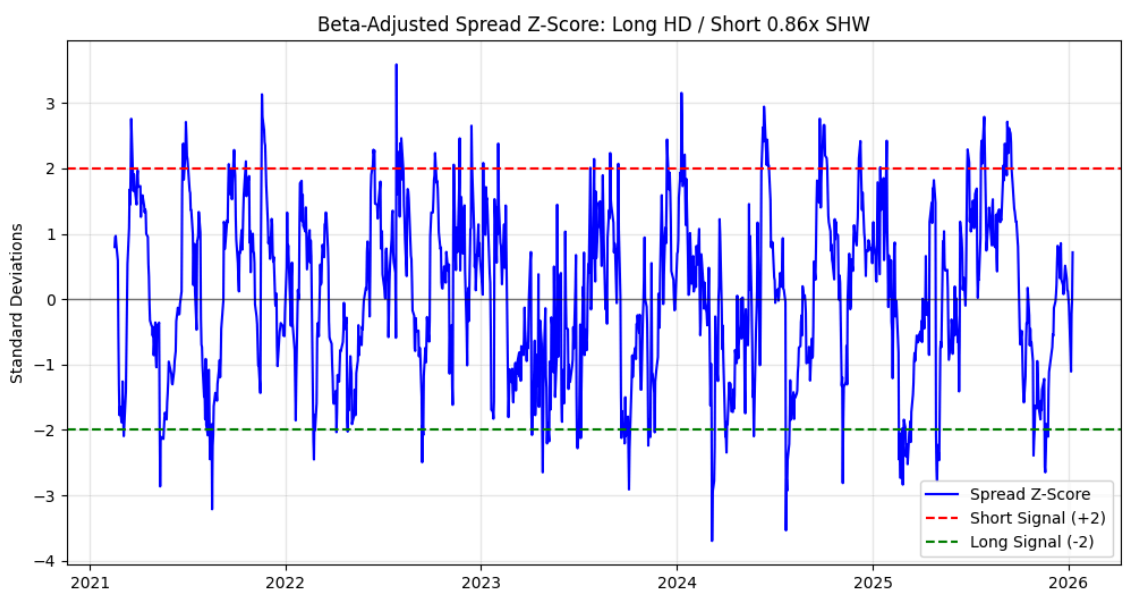
When looking at the graphs of the prices by themselves, there appears to be a very clear similarity in how the stocks move over time.

To better visualize the cointegration between SHW and HD, I divided all of the historical stock prices by the initial day's price for each company and multiplied the resulting historical prices by 100. This allows the historical graphs of both tickers to begin from the same starting point and display their growth and decay relative to their beta values. I also plotted the absolute value of their spread over time to show how it doesn't have drastic growth or decay over time.

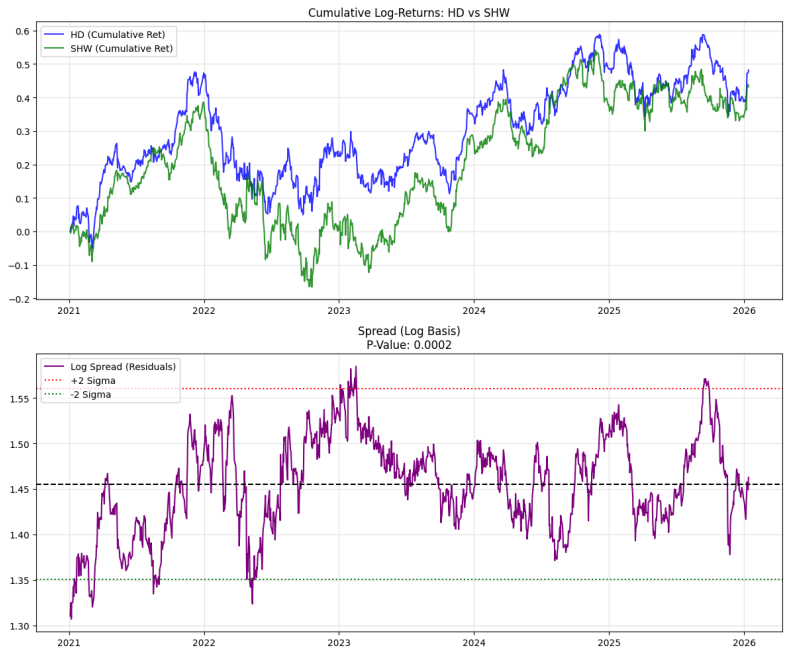


I used the Engle-Granger cointegration test to determine whether the two tickers did in fact have that relationship. The test regresses one graph onto another and checks for the stationarity of the residuals. The p-value for this cointegration test was **0.00324**, providing the cointegration between SHW and HD to be significant.

After calculating a hedge ratio between HD and SHW for one 30-day window, I calculated the rolling z-scores of the spread (HD - beta*SHW) with a window of 30 days and plotted them to show how the spread mostly remains within 2 standard deviations of the mean spread. In the final rolling window, one standard deviation was $5.37, to give a better estimate of the margin for a trade. Beta was 0.86, meaning that if the spread was $10.74 greater than the rolling mean, I would long 0.86 shares of SHW for every share I short of HD to bet on mean reversion.



Another useful graph is the log price series because that gives information about how the spread of the cumulative returns changes over time, since log returns are additive. The cointegration test for the log spread again yielded a significant p-value (0.00017) and the graph of the spread embodied the mean reversion more clearly than the original spread z-score graph. We also adjust for beta here, with a value of 0.76, meaning that for every 1% move of HD, there tends to be a 0.76% move of SHW



Trade Expression

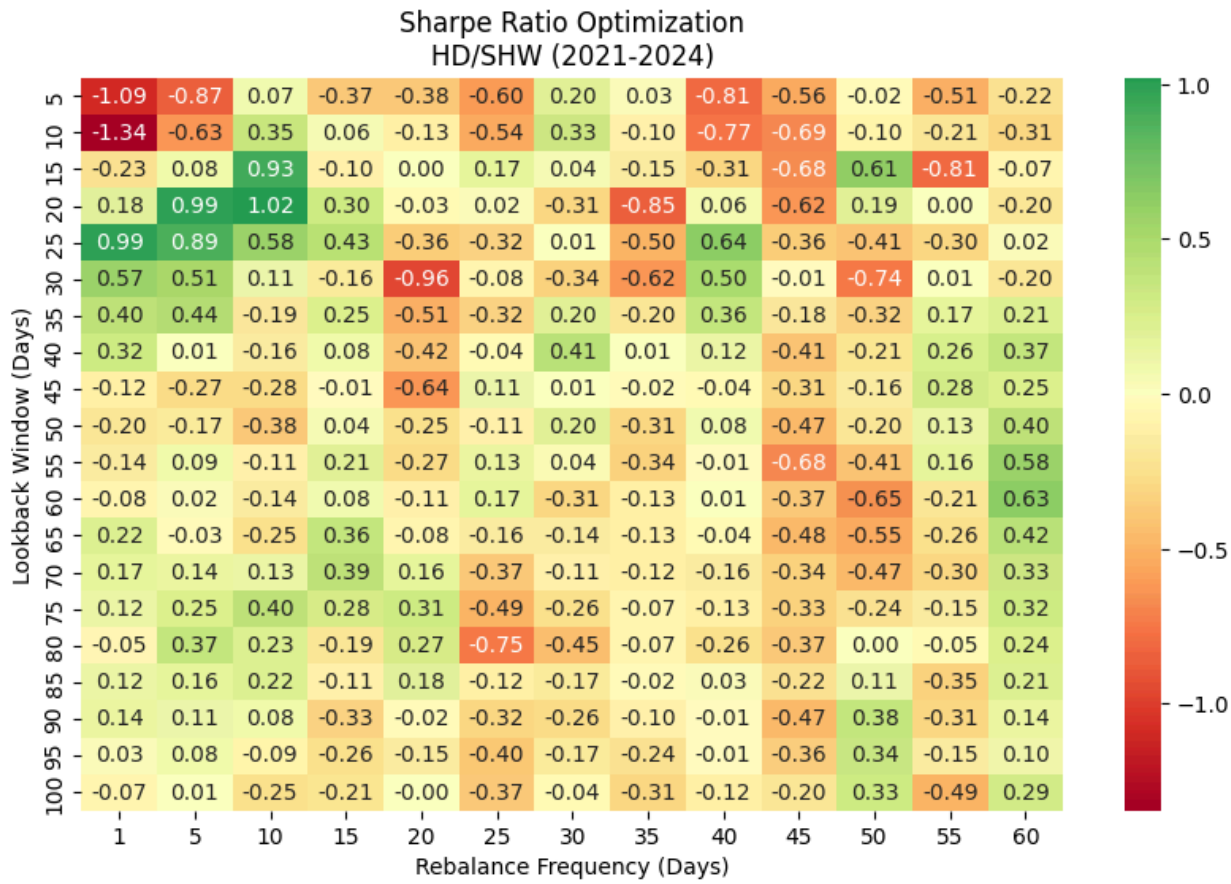
When the rolling z-score of the spread is greater than 2, I will short the spread (short HD, long beta-adjusted SHW). When the z-score is less than -2, I will long the spread (long HD, short beta-adjusted SHW). I will exit the trade at z = +-0.5. On days that I rebalance my variables, I will pay transaction costs to adjust my position since the beta will be changing and I will have to adjust the ratio of HD shares to SHW shares.

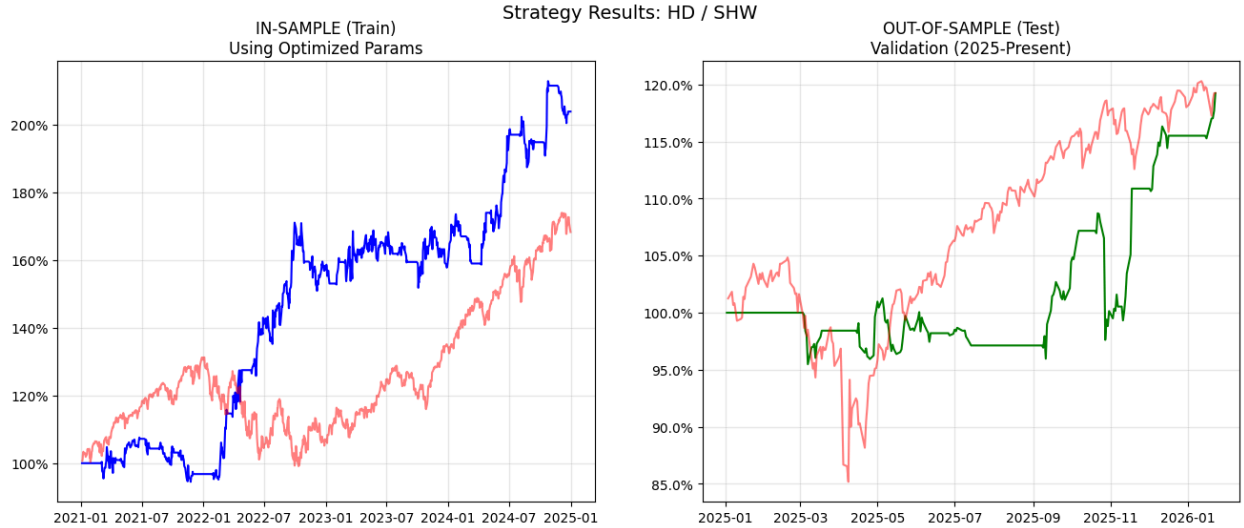
Backtesting & Validation

I optimized for lookback period and frequency: how large my rolling window is to calculate the hedge ratio and mean/standard deviation for a z-score calculation and how often I would adjust these variables. In a frictionless market, it would be ideal to have the highest lookback frequency possible because we would have a perfect understanding of the hedge ratio between both stocks; however, the transaction costs associated with constantly adjusting the hedge ratio would make it more sensible to readjust every so often instead.

When looking back for a window of 20 days every 10 days, we get a sharpe ratio of 1.02. Although, I am concerned about the instability of these parameters because it looks like there isn't too much of a gradient across parameters, it is more spotty. This is likely due to the lower frequency of my trades which gives more weight to trades where I take the wrong direction,

which ruins the Sharpe ratio. However, the EDA has proven the cointegration between these stocks to be significant and there are several high-sharpe combinations of parameters around the one I have chosen.





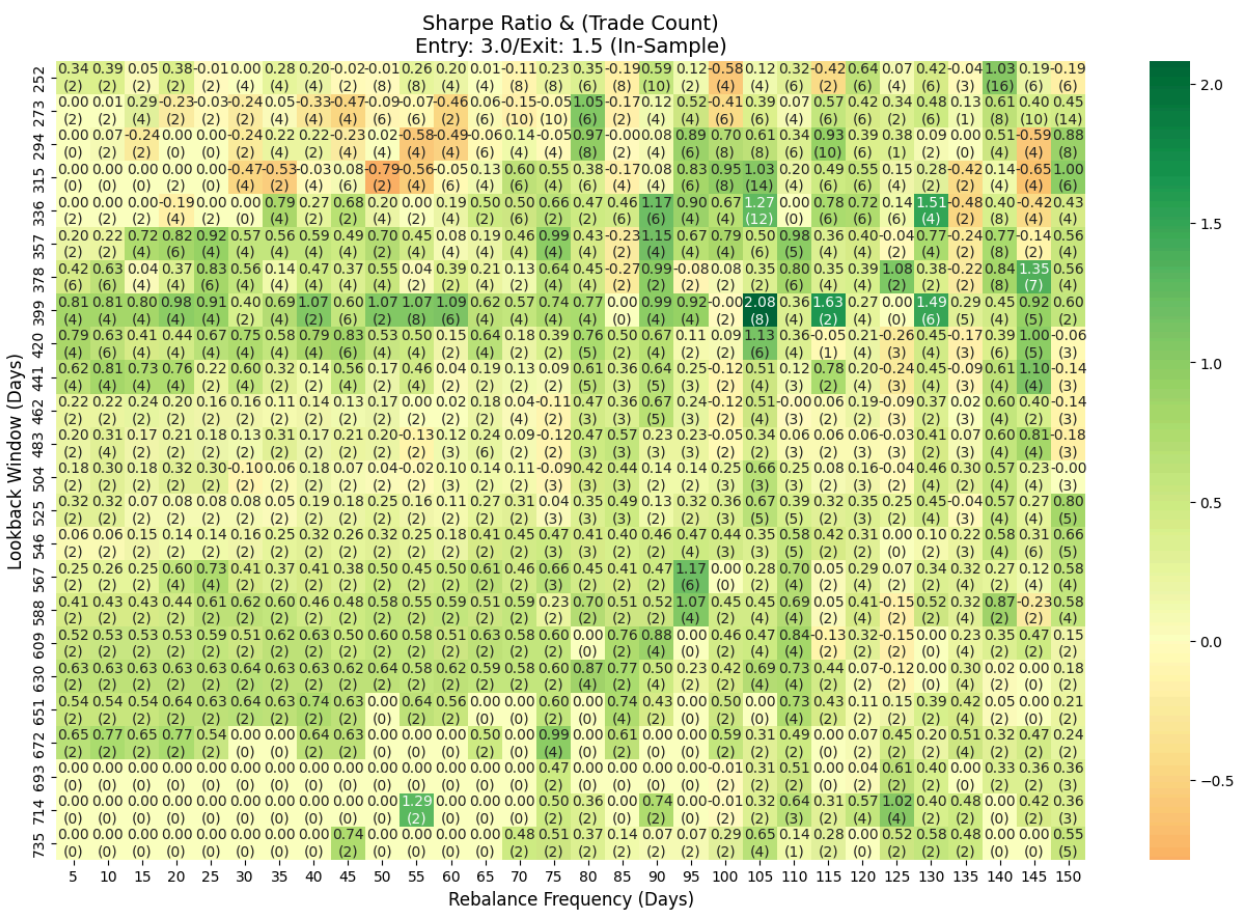
Further considerations

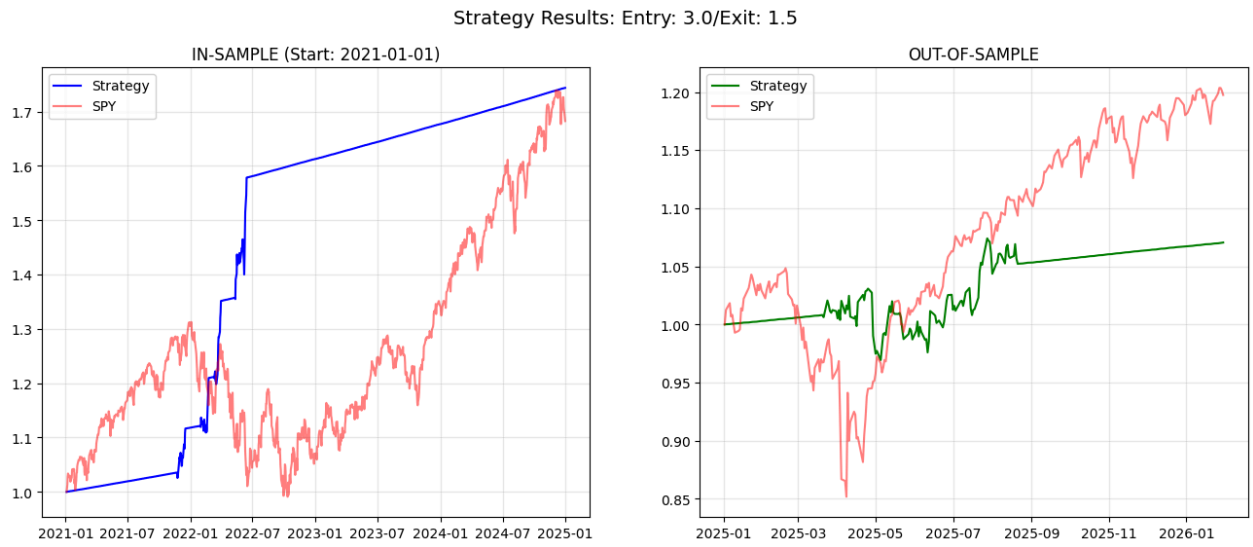
My selection of entry z = 2 and exit z = 0.5 was initially arbitrary so I looked through optimization heatmaps for other combinations of entry and exit z scores and found that my highest sharpe came from a combination of 3 and 1.5 for my entry and exit z scores.

I checked for lookbacks between 1 and 3 years and rebalances between 5 and 150 days since the parameters were very unstable before, likely because of the small amount of data going into the rolling z-score calculation. My maximum sharpe ratio was 2.08 with a lookback of 399 days and rebalance frequency of 105 days using entry and exit z-scores of 3.0 and 1.5. In some other z-score combinations, there were other high-sharpe cells that appeared to have more stable parameters so I do want to further investigate these.

Revised Trade Expression

When the rolling z-score over a lookback period of 399 days of the spread is greater than 3, I will short the spread (short HD, long beta-adjusted SHW). When the z-score is less than -3, I will long the spread (long HD, short beta-adjusted SHW). I will exit the trade at z = +-1.5. I will change my lookback window every 105 days. On the days that I rebalance my variables, I will pay transaction costs to adjust my position since the beta will be changing and I will have to adjust the ratio of HD shares to SHW shares. I collect risk-free on the days that I am not in position due to the low frequency of trades.




Strategy Results: Entry: 3.0/Exit: 1.5
IN-SAMPLE (Start: 2021-01-01)
Strategy
SPY
1.7
1.6
1.5
1.4
1.3
1.2
1.1
1.0
2021-01
2021-07
2022-01
2022-07
2023-01
2023-07
2024-01
2024-07
2025-01
OUT-OF-SAMPLE
Strategy
SPY
1.20
1.15
1.10
1.05
1.00
0.95
0.90
0.85
2025-01
2025-03
2025-05
2025-07
2025-09
2025-11
2026-01